

Structural-Zero Components in Audio Anti-Spoofing: Metric Identifiability Under Component Pooling

Proofs and passages omitted from the ICASSP 2027 version

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Every paragraph below is taken verbatim from the paper’s source before it was shortened to the 4+1-page limit; equation, table and section numbers are those printed in the paper. Every number is recomputed from per-clip scores by `verify.py` in this repository.

1 Theorem 1: statement and full proof

Theorem 1 (collapsibility): *Let T be a ROC summary, continuous, non-decreasing under pointwise dominance and strictly so under strict dominance, on a convex class $\mathcal{R}$ whose image $T[\mathcal{R}]$ is an interval and in which every R is strictly dominated by, and strictly dominates, curves attaining every value of $T[\mathcal{R}]$ above and below $T(R)$. Then T admits an aggregation rule $T_{\text{pub}} = \varphi(T_0, T_a, w_0)$ for all curves and all w_0 if and only if $T = g \circ L$ with g continuous and strictly monotone and $L[\text{ROC}] = c + \int_0^1 \text{ROC} dW$ for a fixed finite non-negative Borel measure W , in which case φ is the quasi-arithmetic mean generated by g^{-1} .*

Proof. Sufficiency. With $T = g \circ L$ and $w_0 + w_a = 1$ the constant is absorbed, so L is affine under mixing; writing $L = g^{-1} \circ T$ and applying g gives the rule with $\varphi(x, y, w) = g(w g^{-1}(x) + (1 - w)g^{-1}(y))$.

Necessity. Fix w and set $x \oplus y = \varphi(x, y, w)$ on $I = T[\mathcal{R}]$. Mixing is idempotent, so $x \oplus x = x$; $\oplus$ is strictly increasing in each argument because mixing preserves pointwise dominance; and it is bisymmetric, since $(A \oplus B) \oplus (C \oplus D)$ and $(A \oplus C) \oplus (B \oplus D)$ are both $w^2 A + w(1 - w)(B + C) + (1 - w)^2 D$. Continuity is inherited from T , and the richness hypothesis carries all four to I^4 . Aczél [1, §6.4] then gives a continuous strictly monotone k_w and $\alpha(w) \in (0, 1)$ with $x \oplus y = k_w^{-1}(\alpha(w)k_w(x) + (1 - \alpha(w))k_w(y))$. Mixing at v after w is mixing at weights $vw, v(1 - w), 1 - v$, so $\{\oplus_w\}$ is mutually bisymmetric, which for quasi-arithmetic means forces a common generator k [2, Chs. 17–18]; idempotence and the same composition then make α multiplicative and continuous, so $\alpha(w) = w$. Hence $L := k \circ T$ is affine on $\mathcal{R}$ and continuous in the sup norm, so it extends to a continuous affine functional on the closed convex hull of $\mathcal{R}$ in $C[0, 1]$; by the Riesz representation theorem every such functional is $c + \int_0^1 \text{ROC} dW$ for a finite signed Borel W , and monotonicity of T forces $W \geq 0$. Take $g := k^{-1}$. $\square$

2 Lemma and Proposition 2

Lemma: *If T admits an aggregation rule φ and $T(R_0) = T(R'_0)$, then $T(wR_0 + (1 - w)S) = T(wR'_0 + (1 - w)S)$ for every $S \in \mathcal{R}$ and every w , because both sides equal $\varphi(T(R_0), T(S), w)$.*

Proposition 2: *The equal error rate, the minimum detection cost and Youden’s index admit no aggregation rule.*

Proof. Let A, B and S join $(0, 0)$ to $(1, 1)$ by straight segments through the interior breakpoints $(\frac{2}{5}, \frac{3}{5})$; $(\frac{3}{11}, \frac{1}{2})$, $(\frac{2}{5}, \frac{3}{5})$; and $(\frac{2}{7}, 1)$. All three are concave, hence equal to their own convex hulls. Curves A and B share an equal error rate of $\frac{2}{5}$ yet differ as curves (AUC $\frac{3}{5}$ against $\frac{34}{55}$), and mixing each with the same S at $w = \frac{1}{2}$ gives $\frac{2}{7}$ and $\frac{3}{11}$; both pooled crossings fall on curve vertices, so they are exact rather than interpolated. A second triple, through $(\frac{1}{3}, \frac{2}{3})$; $(\frac{1}{6}, \frac{1}{2})$, $(\frac{1}{3}, \frac{2}{3})$; and $(\frac{1}{5}, \frac{4}{5})$, shares a minimum detection cost and an index of $\frac{1}{3}$ at equal priors and unit costs, and pools to $\frac{4}{24}$ against $\frac{4}{15}$ and to $\frac{5}{12}$ against $\frac{7}{15}$. Each pair contradicts the Lemma, so no φ exists; on the same curves every weighted area has residual exactly zero. A third triple, in the released code, separates the minimum detection cost at the ASVspoof 5 cost parameters. $\square$

3 Theorem 3: statement and full argument

Theorem 3 (sharp partial identification): *A published table fixing w_0 and EER_{pub} confines the component-only value to*

$$\left[\max \left(0, \frac{\text{EER}_{\text{pub}} - w_0}{w_a} \right), \min \left(1, \frac{\text{EER}_{\text{pub}}}{w_a} \right) \right], \quad (1)$$

and both endpoints are attained, so the interval is the sharp identified set given $(w_0, \text{EER}_{\text{pub}})$.

Write τ^* for the threshold at which the published rates cross, so $\text{FAR}(\tau^*) = \text{FRR}_{\text{pub}}(\tau^*) = \text{EER}_{\text{pub}}$. Evaluating (1) there, with $\text{FRR}_0(\tau^*) \in [0, 1]$ the only freedom, gives $\text{FRR}_a(\tau^*) \in [(\text{EER}_{\text{pub}} - w_0)/w_a, \text{EER}_{\text{pub}}/w_a]$. Because $\mathcal{N}$ is shared the false-acceptance rate is common to both components, so $\text{FAR}_a(\tau^*) = \text{EER}_{\text{pub}}$; since FAR falls and FRR_a rises in τ , the component-only crossing EER_a lies between $\text{FAR}_a(\tau^*)$ and $\text{FRR}_a(\tau^*)$. As $(\text{EER}_{\text{pub}} - w_0)/w_a \leq \text{EER}_{\text{pub}} \leq \text{EER}_{\text{pub}}/w_a$ whenever $\text{EER}_{\text{pub}} \leq 1$, that bracket is contained in (3); intersecting with $[0, 1]$ gives the stated form. Both directions are checked numerically in the released code.

Both endpoints are attained, so (3) is sharp given $(w_0, \text{EER}_{\text{pub}})$: for every sampled value in it, endpoints included, the released code builds an admissible system matching both crossings to 10^{-12} . Conditioning additionally on the other published cells of a particular submission constrains the construction further; there we exhibit a witness for the lower endpoint only, so under that stronger conditioning (3) is an outer bound. Bounding a component functional under a two-component mixture is classical [3]; (3) bounds from a single published scalar. The reference baseline’s environmental row on the evaluation split has $w_0 = 0.5323$ and $\text{EER}_{\text{pub}} = 0.4336$, so (3) returns $[0, 0.9271]$, that is 93% of the range, containing the true 0.2090 along with almost everything else.

4 Passages shortened in the conference version

Applicability signal (paper §3.1)

The official submission format requires a predicted class and three scores, including a score for the `original` class [4]. While this is not an explicit applicability flag, it is a strong signal: on the reference system, this score separates the structural zeros with an AUC of 0.9964.

Acoustic difficulty versus pooling (paper §4.3)

A natural objection is that environmental spoofing is simply harder to detect than speech manipulation, so the higher pooled error reflects the model rather than the protocol. A controlled ablation separates the two.

When evaluated strictly on mixed classes where genuine component annotations exist, the environmental and speech heads achieve comparable EERs of 0.2090 and 0.1906 respectively, a marginal intrinsic difference of only 0.0184.

When structural zeros are reintroduced under the benchmark mapping, the environmental EER rises by 0.2246, whereas the speech EER shifts by merely 0.0087. This allows an exact additive decomposition of the reported 0.2343 performance gap:

$$\Delta_{\text{total}} = \Delta_{\text{pooling}} + \Delta_{\text{intrinsic}} = 0.2159 + 0.0184 = 0.2343, \quad (2)$$

where the net contribution from the pooling convention alone ($0.2246 - 0.0087 = 0.2159$) accounts for over 92% of the observed disparity.

Applicability-conditioned evaluation (paper §4.4)

Table 2 outlines what an external reader can recover from outside the scoring process. We set $\gamma_a = 1 - \sigma(s_o)$, with s_o the submitted `original` score and σ the logistic function, and score each axis over $\{x : \gamma_a(x) \geq \frac{1}{2}\}$, at a precision of 0.999 and a recall of 0.943. The absolute deviation from the component-only value falls by 97% on the environmental axis and by 78% on the speech axis, and sweeping the gate threshold from 0.1 to 0.9 moves the environmental error only between 0.0036 and 0.0096.

Relation to prior work (paper §2)

Relation to prior work: The integral class is not new [5], and characterizations of the same shape exist for confusion counts [6], association measures [7], and fairness auditing under unobserved protected class [8]. Those aggregate over groups that carry their own negatives, and in that configuration even the area stops aggregating: it acquires cross-group terms and can fall outside the range of the group values [9]. Equation (2) instead splits the positive side against a shared negative set. The cross terms vanish, the area returns to the collapsible side, and what is left outside the class is the fixed point. It also separates the setting from selective prediction [10], where the model abstains rather than the protocol pooling inapplicable inputs. Within anti-spoofing, unease about the equal error rate is not new: the t-DCF was introduced because the EER operating point does not reflect application requirements [11], and the t-EER removes its free parameters in a tandem setting [12]. Those lines change the operating point or the cost model while holding the evaluation population fixed. Here the population itself carries rows to which the axis does not apply.

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